Birla Vishvakarma Mahavidyalaya Engineering College
(An Autonomous Institution)
Managed by Charutar Vidya Mandal
IT Department

Mid-1 Examination
Project (4IT31)
Academic Year 2026-27

Group 09

Chess Playing Robotic Arm

Team:
Aashka Shah (23IT403)
Veera Agrawal (23IT431)

Project Definition and Objectives

Definition

A smart chess-playing robotic system that uses computer vision to detect piece movements. The system compares previous and current board states, validates moves using a chess engine, and controls a robotic arm to perform the AI-generated move on the physical board.

Objectives

- Automate physical chess gameplay and help remote chess players.
- Detect chess moves using camera.
- Integrate AI for intelligent move generation.
- Control a robotic arm for piece movement.
- Demonstrate the application of AI and robotics in board games.

Modules

1. **Camera-Based Board Monitoring**

   Detect piece positions using camera

2. **Move Detection Module**

   Compare board states and identify moves

3. **Chess AI Engine**

   Validate moves and generate optimal responses

4. **Coordinate Mapping Module**

   Convert chess coordinates into robotic arm coordinates

5. **Robotic Arm Control Module**

   Execute pick-and-place operations

6. **Game Management Module**

   Maintain game history and board state

System Functionality

1. Player moves a chess piece.
2. Camera captures board changes.
3. System identifies source and destination squares
4. Chess engine validates the move
5. AI calculates the best response
6. Robotic arm executes the move
7. Board state is updated
8. Gameplay continues until completion

Feasibility

1

Technical Feasibility

Uses Raspberry Pi, Arduino, robotic arm, Python, and Stockfish AI with readily available components

2

Economic Feasibility

Moderate cost; hardware components are affordable and easily available online

3

Operational Feasibility

Fully automated system with reliable camera-based move detection and robotic piece movement

4

Future Feasibility

Can be extended with multiplayer support, voice control, automatic board setup,

Tech Stack

Front End

Back End

AI Engine

Database

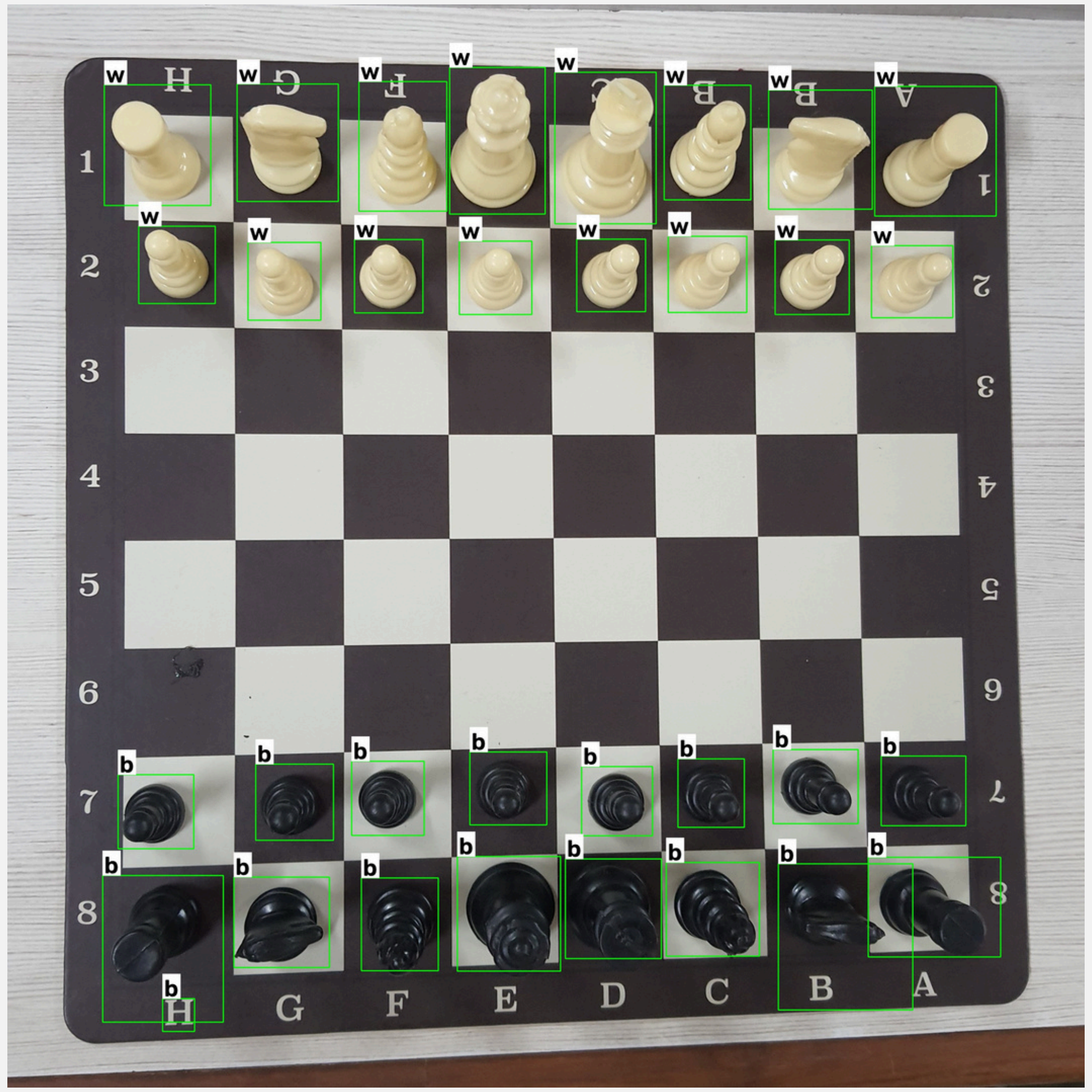
PostgreSQL

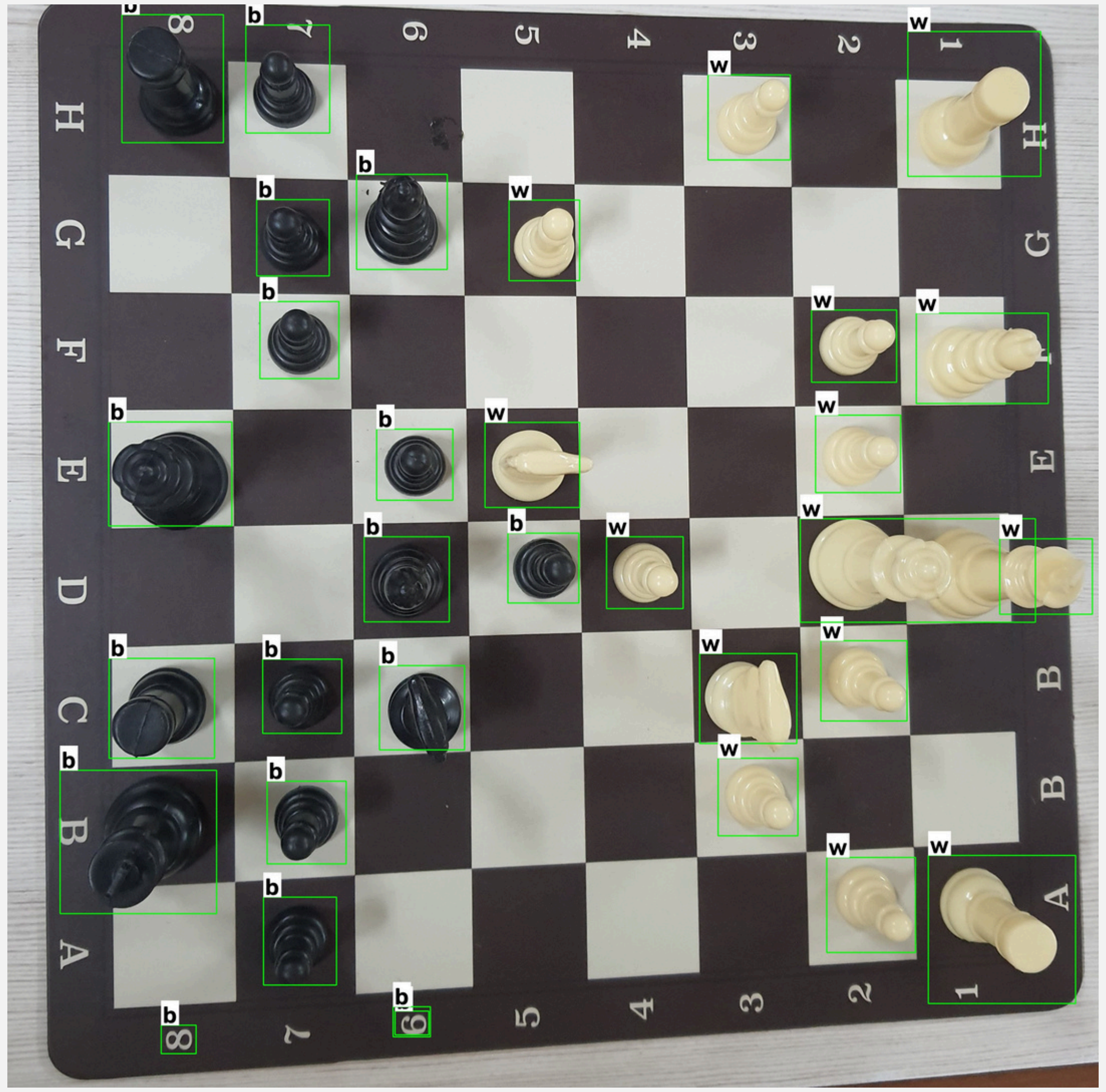
Hardware

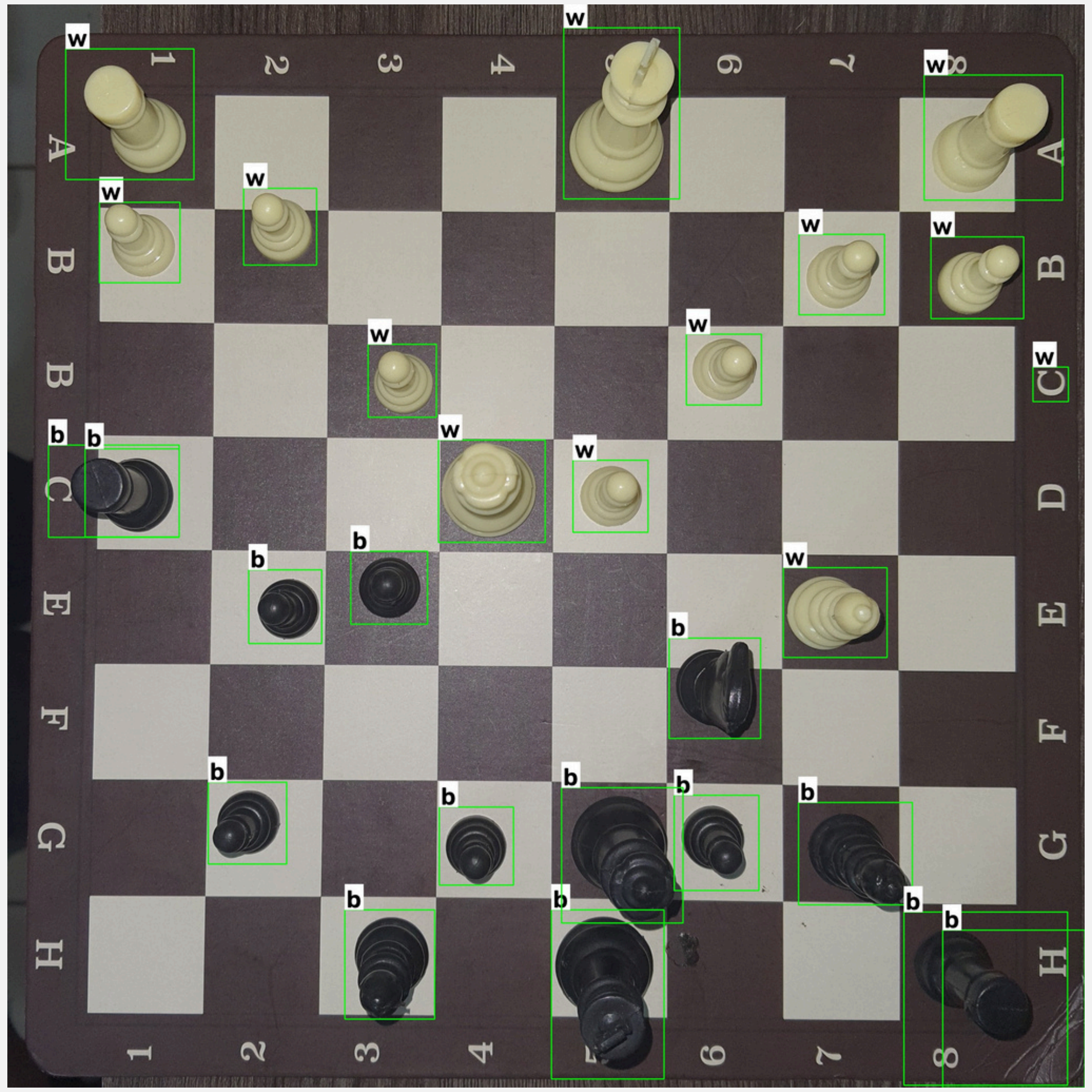
Dataset Roboflow Chess Dataset

Tools

Software Implementation

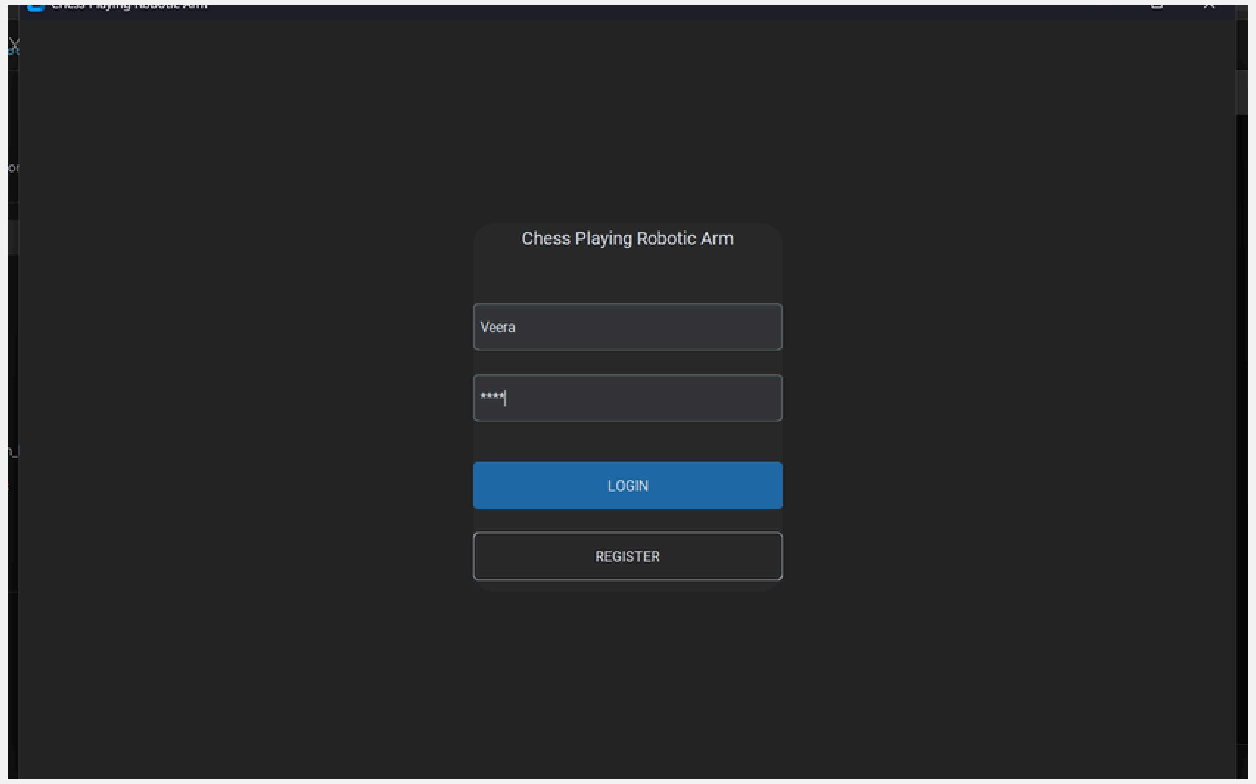


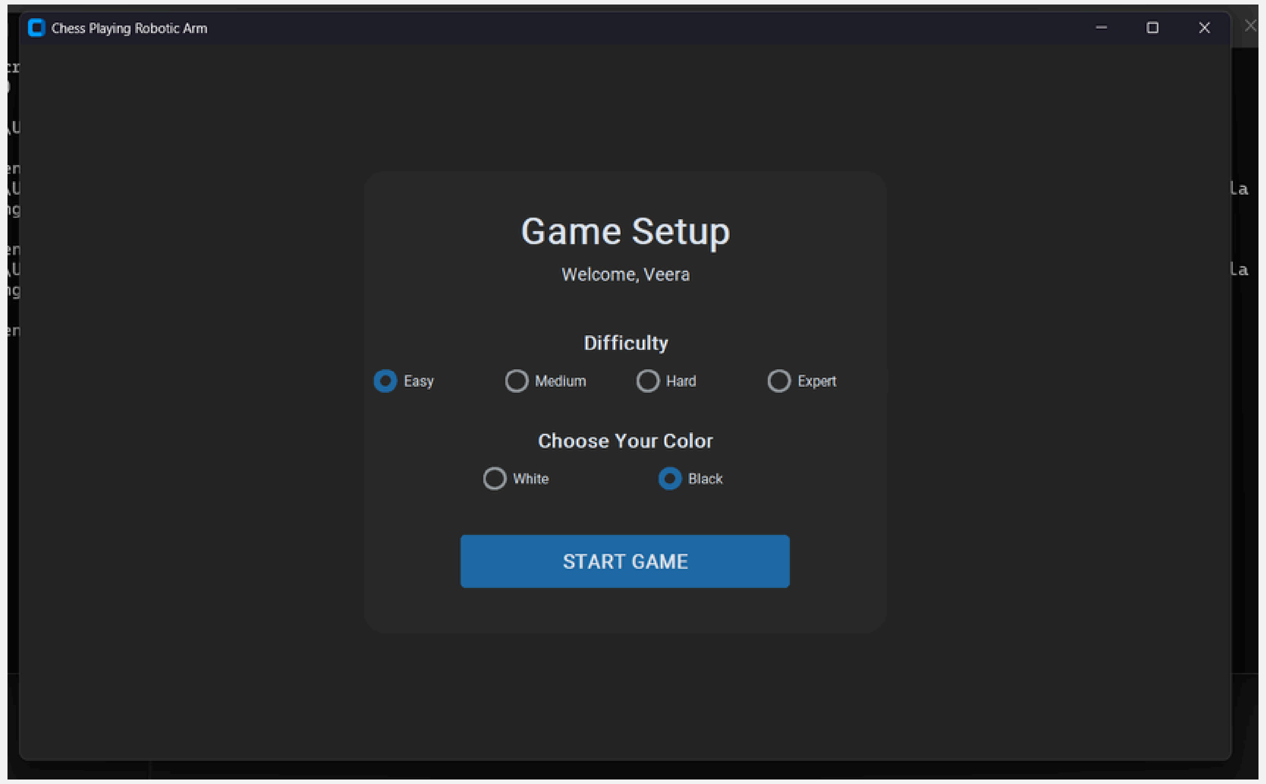


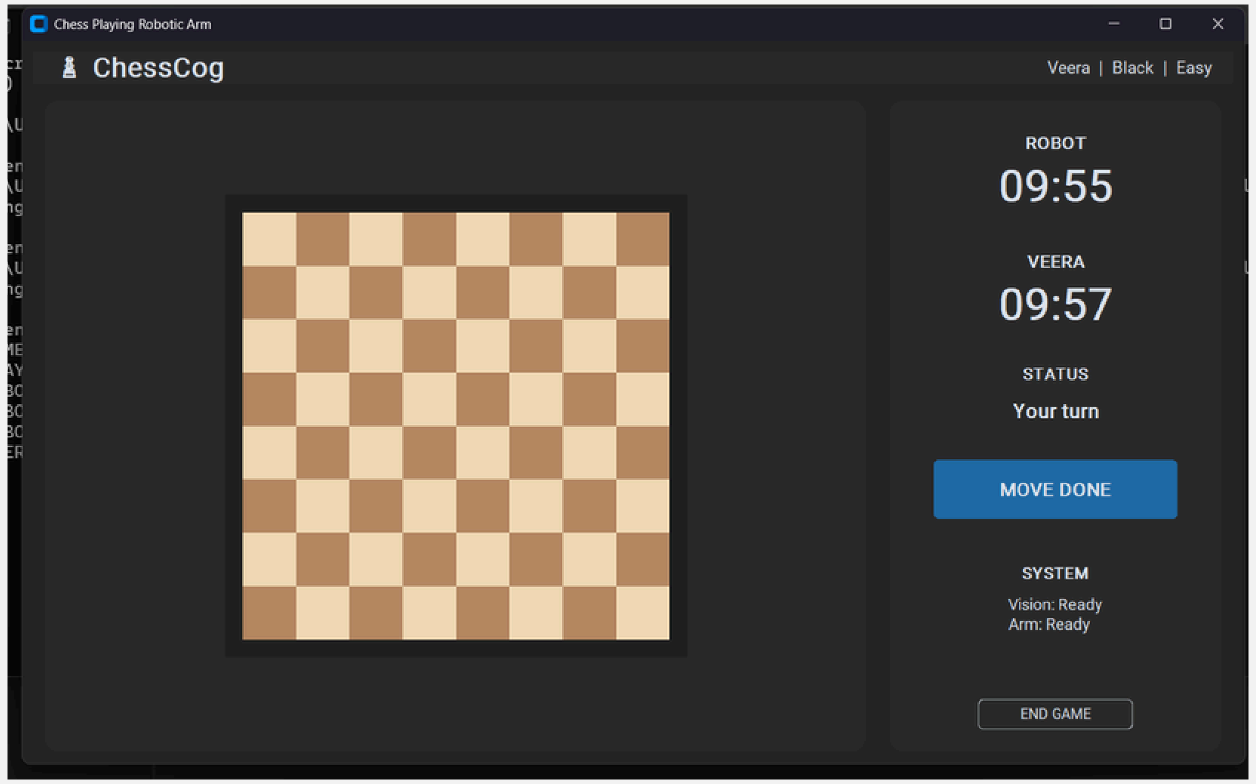


Detects if piece is black or white

UI Implementation



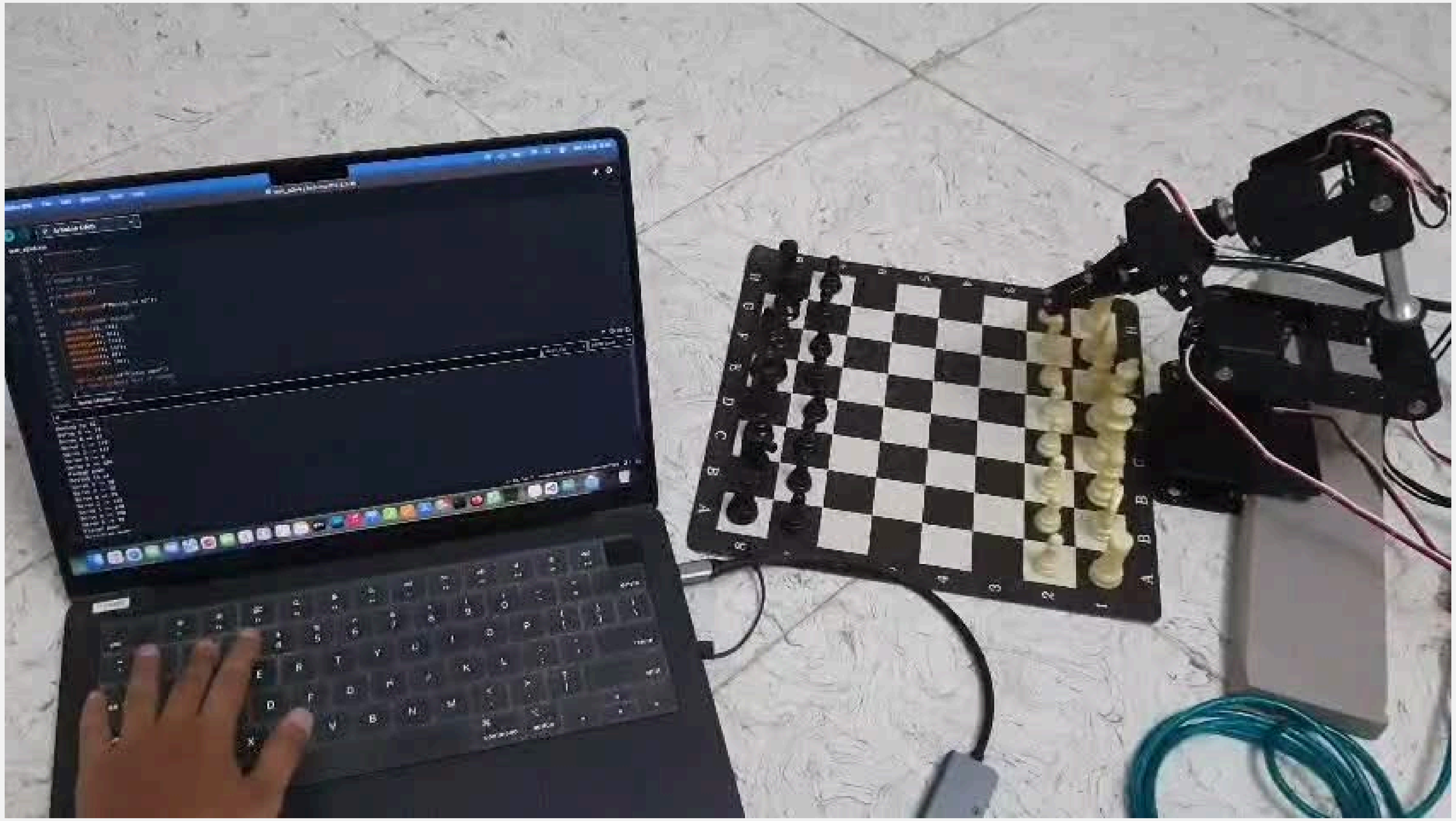




Hardware Implementation

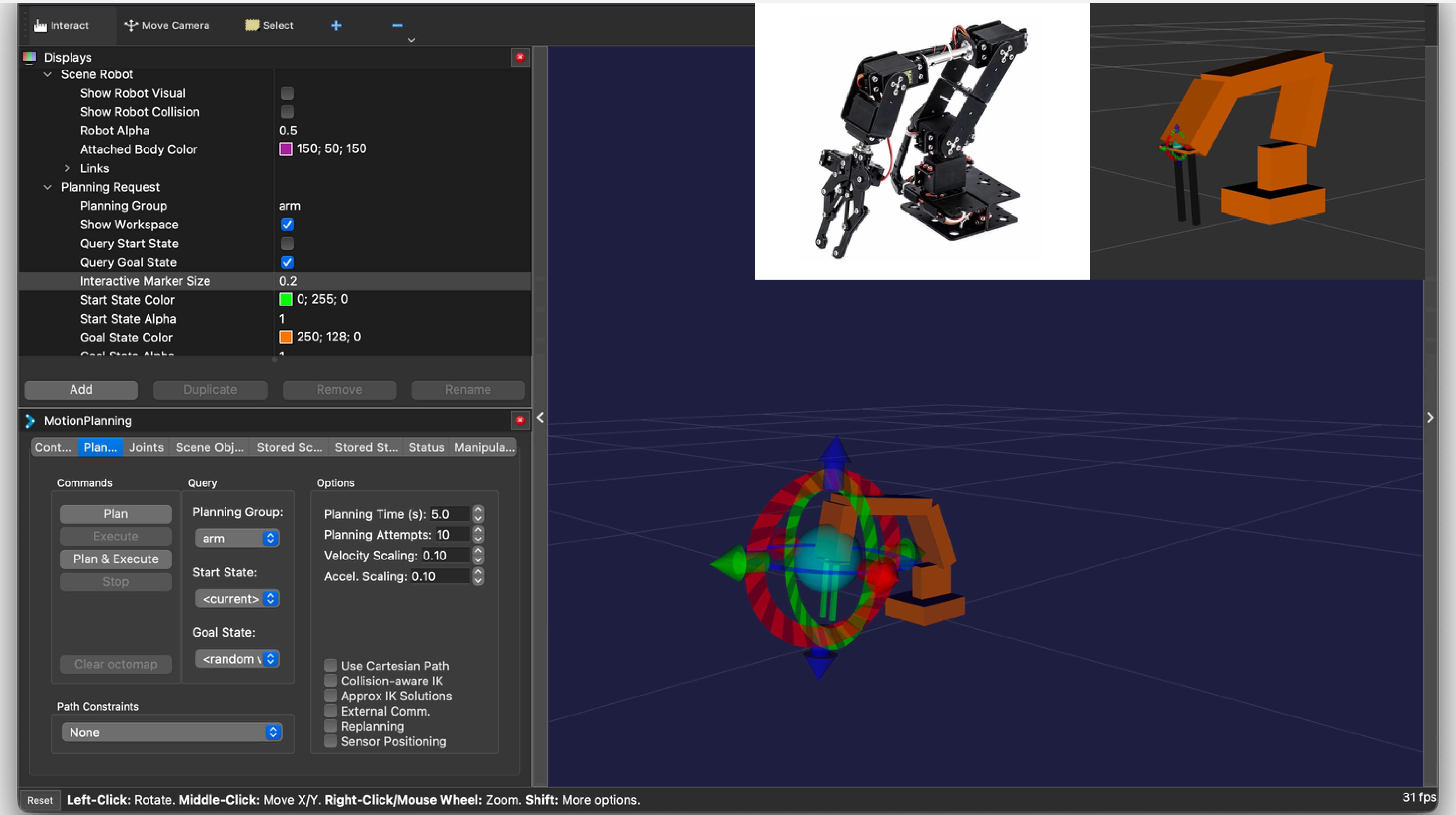
Assembled arm going to initial position on power

Hardware Implementation



Robotic arm move: e2-e4

Hardware Implementation



Arm simulation

Current Goal

1. Achieve pick and place by robotic arm
2. Accuractely detect and recognize chess board and pieces
3. Generate the robotic arm move
4. Play move by arm

Future Goal

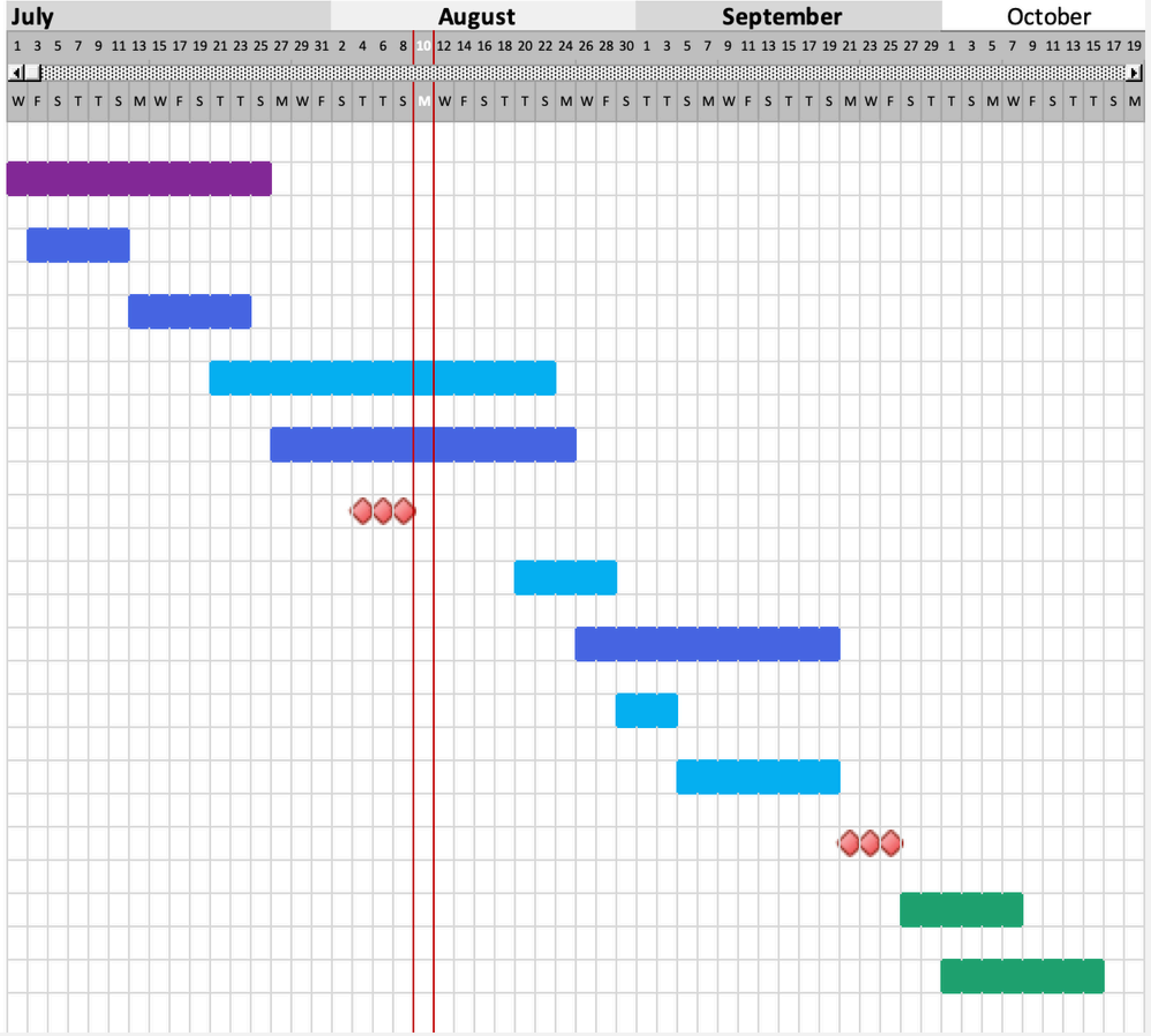
1. Enable robotic play a complete match
2. Handle special cases like enpassant, pawn promotion, castling
3. Improve accuracy and reduce move time
4. Develop this project as a commercial product

Implementation Status

PROJECT: Chess Playing Robotic Arm

Project start date: 01/07/26

Milestone description	Progress	Start	Days
Requirement Analysis & Planning	100%	01-Jul-26	25
Hardware Selection & Procurement	100%	03-Jul-26	10
Hardware Setup	100%	13-Jul-26	12
Piece Recognition & Move Detection	40%	20-Jul-26	35
Coordinate Mapping & Motion Planning	30%	27-Jul-26	30
Mid Semester 1 Examination	-	03-Aug-26	6
Camera Calibration & Board Monitoring	0%	20-Aug-26	10
Robotic Arm Control	15%	25-Aug-26	27
Chess AI Integration	0%	30-Aug-26	5
Game Management & User Interface	50%	05-Sep-26	15
Mid Semester 2 Examination	-	21-Sep-26	6
System Integration	0%	27-Sep-26	11
Testing & Optimization	0%	01-Oct-26	15



Links to Documentation

Progress Report 1

Progress Report 2

Project SRS

THANK YOU